SDTM and SDTMIG Conformance Rules

Version 2.0 (Final)

Developed by the
SDS Study Data Tabulation Model (SDTM)
Conformance Rules Subteam

Notes to Readers

- This is the final version of the conformance rules for the Study Data Tabulation Model (SDTM) and SDTM Implementation Guide (SDTMIG), which includes rules for all SDTM versions through SDTM v2.0 and for all SDTMIG versions through SDTMIG v3.4.

Revision History

Date	Version	Description of Changes
2021-11-29	2.0 Final	• Removed "distinct" from "4 distinct categories" in Appendix A • Changed 4a. to "FDA Validator Rules describe the validation rules used by the FDA" • added 4c "PointCross' eDataValidator validator rules describe the validation rules used by PointCross' eDataValidator." • Changed "For example" to "Following are some examples of validator rule sets; others are also available:" in 4. • Changed the order of the Condition and Rules columns so that the condition comes before the rule • Updated the order of the columns to match the order on the Excel spreadsheet. Also updated column names to be the same as the spreadsheet.
2020-05-07	1.1 Final	
2016-12-12	1.0 Final	

See Appendix B for Representations and Warranties, Limitations of Liability, and Disclaimers.

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1 Introduction

This guide was developed by the CDISC Submission Data Standards (SDS) Study Data Tabulation Model (SDTM) Conformance Rules subteam to explain the methodology used to create SDTM conformance rules and the logic applied to determine which implementation guidance should be translated into rules.

1.1 Purpose

The purpose of this guide is to provide methodology for the identification and classification of implementation text that constitutes a rule definition.

1.2 Scope

SDTM conformance rules have an explicit basis in text from the SDTM document, the SDTM Implementation Guide (IG), and the SDTMIG: Associated Persons.

The following are considered to be outside the scope of SDTM conformance rules:

- Rules that are not supported by a definitive statement in the underlying documents
- Business rules
- Data validation logic that is not specifically linked to SDTM implementation guidance

SDTM conformance rules are identified and classified regardless of the perceived programmability of the rule.

For each identified SDTM conformance rule, a logic statement is provided, consisting of scoping attributes (e.g., applicable domains); the rule itself; and, if applicable, a conditional statement. Unless otherwise specified, it is implicit in rule logic that it applies to a given subject or record.

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2 Methodology for the Identification of SDTM Conformance Rules

The following table provides descriptions of the columns in the SDTM Conformance Rules template.

Rules Template Column	Description
Rule ID	Identifier used to uniquely identify a rule
SDTMIG Version	Indicates the IG version to which the rule applies. This allows subsetting of the rules across multiple documents (e.g., Version 3.2 identifies all rules from SDTM v1.4, SDTMIG v3.2, and SDTMIG-AP v1.0).
Rule Version	Each rule is versioned. A value of 1 indicates the rule is largely unchanged from the SDTMIG v3.2 ruleset as originally published, or is a new rule identified in this release. Version 2 indicates a rule published in v1 of the rules has been updated. A rule is considered updated if there is a change to the Class, Domain, Variable, Rule, or Condition (e.g., changes that affect implementation).
Class	The Observation Class to which the rule applies is indicated using an abbreviation: • TDM – Trial Design Model Datasets • SPC – Special Purpose Class • FND – Findings Class • EVT – Events Class • INT – Interventions Class • REL – Relationship Datasets • AP – Associated Persons • ALL – All Classes If the rule applies to multiple classes, each observation class is listed separated by commas. If the rule applies to all classes, ALL is used.
Domain	The domain to which the rule applies is indicated using the 2-character domain abbreviation. Exceptions: RELREC, SUPP--. If the rule applies to multiple domains, each domain abbreviation is listed separated by commas. If the rule applies to all domains, ALL is used.
Variable	Indicates the variable name to which the rule applies. If the rule applies to a variable across multiple domains, the variable name is presented with dashes in place of the domain abbreviation (e.g., --STDY, --ENDY). If the rule applies to all variables, ALL is used.
Condition	If a rule is applied only in a conditional set of circumstances, the condition is defined in this column. The following are guidelines for composing condition statements: • It is not necessary to preface the condition with "*If*...." This is implicit in the fact that it is a condition. • Any reference to a variable in a domain other than the object variable's domain is in the form "Domain.Variable" (e.g., TA.ARM, DM.SEX). • Multiple parts of the condition should be separated by standard logical operators (e.g., AND, OR, AND NOT). Parentheses are allowed to help clarify when multiple logical operators are used in a single condition. • Controlled terminology or terminology taken from SDTMIG should be used to describe conditions or requirements in a standard way. The following terms and definitions should be used to define rule conditions: • *First*: The first instance (e.g., date); implies sorting • *Last*: The last instance (e.g., date); implies sorting • *Unique:* Only occurrence of value within defined scope • *Exists:* Object is present (e.g., dataset, variable); does not imply that object is populated with data • *Null:* Object does not contain data • *One-to-one:* Object in isomorphic relationship with another object (i.e., value pair is unique)
Rule	A concise and unambiguous statement of the conformance principle to be applied. Only 1 principle is stated per rule. If the rule is applied only when a specific condition is met, then the condition is specified in the Condition column. The following are guidelines for composing rule statements: • All rules are stated uniquely at the highest level of hierarchy possible. • Any reference to a variable in a domain other than the object variable's domain is in the form "Domain.Variable" (e.g., TA.ARM, DM.SEX).

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<table>
<tr><th>Rules Template Column</th><th>Description</th></tr>
<tr><td></td><td>
<ul>
<li>If a variable should be populated with a selection from a discrete list described in guidance but not in CDISC Controlled Terminology, then the syntax should be “Variable in (value1, value2…).” Enclosing the individual values in the list within quotation marks is not recommended unless the values contain special characters such as commas.</li>
<li>If a variable should be populated with a selection from a discrete list described in CDISC Controlled Terminology, then the syntax should be “Variable in {CT List Name}.” Note the use of braces instead of parentheses.</li>
<li>When a variable should not contain a value, use the keyword “null” (e.g., “AGETXT = null”), rather than phrases such as "is missing," "equals blank," or "should not be populated."</li>
<li>Logical operators should be used in place of phrases such as "less than or equal to," "not greater than," "should equal," and so on. This is to enforce a concise and unambiguous rendering of the rule.</li>
<li>The term "one-to-one" is used to identify that an object has an isomorphic relationship with another object (i.e., value pair is unique).</li>
</ul>
The following logical operator substitutions should be used:
<table>
<tr><th>Use:</th><th>Instead of:</th></tr>
<tr><td>=</td><td>Should equal, must equal, equals</td></tr>
<tr><td><=</td><td>Less than or equal to</td></tr>
<tr><td><</td><td>Less than</td></tr>
<tr><td>></td><td>Greater than</td></tr>
<tr><td>>=</td><td>Greater than or equal to</td></tr>
<tr><td>In</td><td>Must be defined in, must include</td></tr>
<tr><td>^, not</td><td>Not; general negation operator</td></tr>
</table>
</td></tr>
<tr><td>Document</td><td>Indicates with which document the rule is associated (e.g., IG v3.2, AP v1.0).
If multiple documents are identified as the basis for a rule, each will be listed, separated by "|". For example, "IG v3.2|Model v1.4" indicates both the SDTMIG and the Model document.</td></tr>
<tr><td>Section</td><td>Indicates the document section/subsection that prescribes the rule. In cases where the rule is prescribed in multiple sections, only the first instance will be included.
If multiple sections are identified, each will be listed, separated by "|". If the sections relate to more than 1 document, the document will be included as a prefix, followed by the associated section enclosed in brackets. For example, "IG v3.2[2.5]|Model v1.4[3.2.22]" indicates IG v3.2, Section 2.5 and Model v1.4, Section 3.2.22.</td></tr>
<tr><td>Item</td><td>Indicates the item within a section/subsection
If multiple items are identified, each will be listed, separated by "|". If the items relate to more than 1 document or section, the document and section will be included as a prefix, followed by the associated sub-item enclosed in brackets. For example,
<ul>
<li>"IG v3.2[2.5]|Model v1.4[3.2.22]" indicates IG v3.2, Section 2.5 and Model v1.4, Section 3.2.22.</li>
<li>"IG v3.2[6.2 AE][Specification]|Model v1.4[2.2.2][Table 2.2.2: --BDSYCD]" indicates:
<ul>
<li>IG v3.2, Section 6.2 AE, Specification Item</li>
<li>Model v1.4, Section 2.2.2, Table 2.2.2: --BDSYCD Item</li>
</ul></li>
</ul></td></tr>
<tr><td>Cited Guidance</td><td>Cited text from the document as indicated in the cited document section on which the rule was based
If multiple guidance citations are identified, each will be listed, separated by "|". If the items relate to more than 1 document, section, and/or item, this information will be included as a prefix, followed by the associated sub-item enclosed in brackets.
For example,
IG v3.2[6.2 AE][Specification][IG: Dictionary derived. Code for body system or organ class used by the sponsor from the coding dictionary (e.g., MedDRA). When using a multi-axial dictionary such as MedDRA, this should contain the SOC used for the sponsor's analyses and summary tables which may not necessarily be the primary SOC.]|Model v1.4[2.2.2][Table 2.2.2: --BDSYCD][Model:MedDRA System Organ Class code corresponding to --BODSYS assigned for analysis.]
indicates:
<ul>
<li>Cited Guidance IG v3.2, Section 6.2 AE, Specification Item:
Dictionary derived. Code for body system or organ class used by the sponsor from the coding dictionary (e.g., MedDRA). When using a multi-axial dictionary such as MedDRA, this should contain the SOC used for the sponsor's analyses and summary tables, which may not necessarily be the primary SOC.</li>
<li>Model v1.4, Section 2.2.2, Table 2.2.2: --BDSYCD Item</li>
</ul></td></tr>
</table>

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Rules Template Column	Description
	MedDRA SOC code corresponding to --BODSYS assigned for analysis.
Release Notes	Free-text description identifying anything of note related to the rule. For example, if a rule has been updated (Version 2), the details of the update are noted. Release notes are also provided if non-impactful changes were made (e.g, the cited guidance was updated or reformatted).

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3 Appendices

Appendix A: Definitions of Rule Sets

There are several types of rule sets surrounding SDTM. These rules can be divided into 4 categories:

1. **Conformance Rules:** Conformance rules are created and maintained by CDISC. Conformance rules describe the criteria that must be met to be in compliance with the CDISC standard.
2. **Business Rules:** Business rules are created by a specific organization to describe the criteria that should be met to allow for the deliverable to be useful in the conduct of normal business practices. For example, US Food and Drug Administration (FDA) Business Rules describe criteria that should be met in order for datasets to be utilized internally for FDA business practices such as submission review.
3. **Technical Rejection Criteria for Study Data:** eCTD Technical Rejection Criteria describe the minimum requirements for eCTD submissions to be accepted by the agency at the gateway.
4. **Validator Rules:** Validator rules are rule sets utilized by validation tools. Each validator can have its own set of validation rules. Following are some examples of validator rule sets; others are also available:
    a. FDA validator rules describe the validation rules used by the FDA.
    b. Pinnacle 21 validator rules describe the validation rules used by the Pinnacle 21 validator tool.
    c. PointCross' eDataValidator validator rules describe the validation rules used by PointCross' eDataValidator.

Rule	Owner	Published Location
Conformance Rules		
SDTM	CDISC	https://www.cdisc.org/standards/foundational/sdtmig
Business Rules		
FDA Business Rules	FDA	https://www.fda.gov/ForIndustry/DataStandards/StudyDataStandards/default.htm
Technical Rejection Criteria for Study Data		
eCTD Technical Rejection Criteria for Study Data	FDA	https://www.fda.gov/drugs/developmentapprovalprocess/
Validator Rules		
FDA Validator Rules	FDA	https://www.fda.gov/ForIndustry/DataStandards/StudyDataStandards/default.htm

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